1. Consider the following simplex tableau:

Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS
max Z	6	c	3	d	0	4	-12
a	0	e	2	f	1	6	5
b	1	1	-2	g	-2	4	2

a	x_4	b	x_2	c	0
d	0	e	0	f	1
g	0				

(i) (7p) Find all unknowns and write the results in the table on the right.

(ii) (8p) The simplex tableau given above indicates ______ solution(s).

A. a unique optimal B. alternate optimal C. an infeasible D. an unbounded

2. (5p) Given the primal and dual problems below:

$$\text{Primal:}\quad \begin{array}{rcl} \max f(x) & = & cx^T \\ \text{s.t. } Ax & \leq & b \\ x & \geq & 0 \end{array} \qquad \text{Dual:}\quad \begin{array}{rcl} \min g(y) & = & b^T y \\ \text{s.t. } A^T y & \geq & c \\ y & \geq & 0 \end{array}$$

where A is an $m \times n$ matrix, b and c are vectors in $\mathbb{R}^m$ and $\mathbb{R}^n$, respectively. If $\bar{x}$ is an optimal solution to the primal problem and $\bar{y}$ is an optimal solution to the dual problem, then which of the following statements is (are) true?

(1): $f(\bar{x}) = g(\bar{y})$ (2): $f(\bar{x}) = f(\bar{y})$ (3): $g(\bar{x}) = g(\bar{y})$ (4): $g(\bar{x}) = f(\bar{y})$

A. (1) only B. (2) and (3) C. (1) and (4) D. All of them E. None

3. Construct the initial feasible solution to the given transportation problem by the given method and report the corresponding objective function value (OFV).

A. (5p) Northwest corner:

2 10	3	1	~~10~~ 0
5 10	4 25	8 0	~~35~~ ~~25~~ 0
5	6	8 25	~~25~~ 0
~~20~~ ~~10~~ 0	~~25~~ 0	~~25~~ 0	

OFV = 2*10+5*10+4*25+0*8+8*25=370

B. (5p) Least-cost:

2	3	1 10	~~10~~ 0
5 10	4 25	8	~~35~~ ~~10~~ 0
5 10	6	8 15	~~25~~ ~~15~~ 0
~~20~~ ~~10~~ 0	~~25~~ 0	~~25~~ ~~15~~ 0	

1 2 3

OFV=

1*10+5*10+4*25+5*10+8*15=330

C. (5p) VAM:

2	3	1 10	10
5 10	4 25	8	35
5 10	6	8 15	25
20	25	25	

OFV=

4. (5p) Consider a primal problem and its dual. Which of the below statement(s) is(are) correct?

i. Unconstrained variables in the primal appear as equality constraints in the dual.

~~ii.~~ Equality constraints in the primal appear as equality constraints in the dual.

~~iii.~~ $\geq$ constraints in the dual always appear as $\geq$ constraints in the primal.

~~iv.~~ For feasible solutions; the objective values of the dual are always greater than the objective values of the primal.

A. Only i B. Only ii C. Only iii D. ii and iii E. iii and iv

5. (10p) Consider the problem and its optimal

$$\begin{aligned}
\min z = \quad & 4x_1 + 2x_2 - x_3 \\
\text{s.t.} \quad & x_1 + x_2 + 2x_3 \geq 3 \\
& 2x_1 - 2x_2 + 4x_3 \leq 5 \\
& x_1, x_2, x_3 \geq 0
\end{aligned}$$

Optimal Duals $= \begin{bmatrix} 2 & -1 \end{bmatrix} \begin{bmatrix} 1/2 & -1/4 \\ 1/4 & 1/8 \end{bmatrix}$ → y_1 ve y_2 nin bağlı olduğu R ve S_2'nin tablo sütunları

original objective function values of basic variable

$\left(1 - \frac{1}{4}, \frac{-1}{2} - \frac{1}{8}\right) = \left(\frac{3}{4}, \frac{-5}{8}\right)$

Basic	x_1	x_2	x_3	S_1	R	S_2	RHS
z	$-9/2$	0	0	$-3/4$	$3/4 - M$	-5/8	-7/8
x_2	0	1	0	$-1/2$	$1/2$	$-1/4$	$1/4$
x_3	$1/2$	0	1	$-1/4$	$1/4$	$1/8$	$11/8$

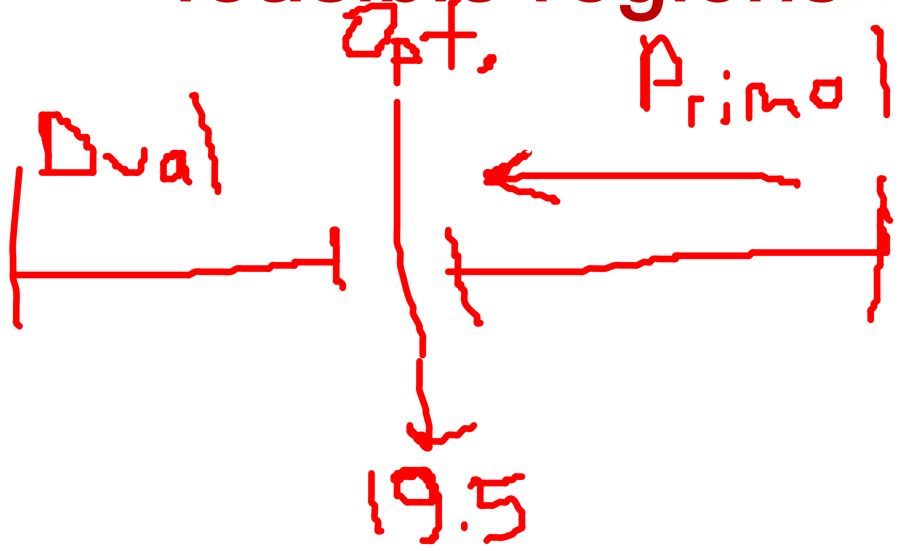
Accordingly, what are the optimal dual values of the problem?

A. $(y_1, y_2) = (-3/4, 5/8)$ B. $(y_1, y_2) = (-5/8, 3/4)$ C. $(y_1, y_2) = (3/4, -5/8)$

D. $(y_1, y_2) = (5/8, -3/4)$ E. $(y_1, y_2) = (3/4, 5/8)$

6. (10p) Consider the problem:

Bu soruda dual simplex yapacağız.



$$
\begin{aligned}
\min z = {} & 15x_1 + 12x_2 \\
\text{s.t.} \quad & 2x_1 + 2x_2 \geq 3 \\
& 2x_1 + 4x_2 \leq 5 \\
& x_1, x_2 \geq 0
\end{aligned}
$$

What can be the objective function value of one of the feasible solutions of the dual problem?

A. 18   B. 20   C. 22   D. 24   E. 26

7. (5p) Which of the following statements is correct?

Primal'in amacı optimal'e ulaşmaktır. Ama dual optimal ve infeasible koşullarıyla başlayıp feasible sonuca ulaşmayı amaçlar(Bu süreçte zaten optimaldir).

i. In the primal simplex method, the LP problem starts with an optimal solution.

ii. In the dual simplex method, the LP problem starts with a solution that is not optimal.

iii. In the dual simplex method, the condition that stops the iterations is the optimality condition.

iv. The primal simplex method cannot be applied to some LP problems.

Primal koşulları: 1-Tüm koşullar nonnegative RHS ile eşit olmalıdır. 2- Her değişken nonnegative olmalıdır.

v. All LP problems can be solved with the dual simplex method.

8. (5p) In dual simplex method; which of the below statements are correct?

~~i.~~ In the starting tableau, the basic variable coefficients in the objective row must be positive.

~~ii.~~ According to the dual feasibility condition, the leaving variable is selected among basic variables with the largest absolute value. The most negative seçilir.

iii. In the dual optimality condition, the entering variable is determined according to the minimum ratio test performed over the negative constraint coefficients where the leaving variable row of the tableau intersects with the columns of the non-basic variables.

iv. In each iteration, the leaving variable is determined at first.

A. Only ii B. Only iv C. ii and iv D. iii and iv E. ii, iii and iv

9. (10p) Solve the given LP with dual simplex algorithm:

$$\min Z = 2x_1 + 3x_2$$

$$\begin{aligned} \text{s.t. } 2x_1 - x_2 - x_3 &\geq 3 \\ x_1 - x_2 + x_3 &\geq 2 \\ x_1, x_2, x_3 &\geq 0 \end{aligned}$$

9-)

$\min z = 2x_1 + 3x_2$

$-2x_1 + x_2 + x_3 \leq -3$

$-x_1 + x_2 - x_3 \leq -2$

$\left.\begin{aligned} &\min z = 2x_1 + 3x_2 \\ &-2x_1 + x_2 + x_3 + S_1 = -3 \\ &-x_1 + x_2 - x_3 + S_2 = -2 \end{aligned}\right\}$ optimal and infeasible $\Rightarrow$ table

Basic	x_1	x_2	x_3	S_1	S_2	Sol
Z	-2	-3	0	0	0	0
S_1	-2	1	1	1	0	-3 → most negative
S_2	-1	1	-1	0	1	-2
min ratio	1	–	–	X	X	
Z	0	-4	-1	-1	0	3
x_1	1	-0.5	-0.5	-0.5	0	1.5
S_2	0	0.5	-1.5	-0.5	1	-0.5
min ratio	X	–	$\frac{2}{3}$	2	X	
Z	0	-13/3	0	-2/3	-2/3	10/3
x_1	1	-2/3	0	-1/3	-1/3	1.5
x_3	0	-1/3	1	1/3	-2/3	1/3

$S_1 \to$ Le.V.
$x_1 \to$ E.V

L.V $\to S_2$
E.V $\Rightarrow x_3$

$-\frac{1}{6} + -\frac{1}{2}$

$-\frac{2}{3}$

$\frac{1}{6} - \frac{1}{2}$

$-\frac{2}{6} = -\frac{1}{3}$

$x_1 = 1.5$, $x_3 = 1/3$ $x_2, S_1, S_2 = 0$

$Z = 10/3$

10. (20p) Obtain the optimal solution(s) of the following transportation problem.

13	4	9	200
10	5	8	175
12	7	6	75
250	100	150	

3600

13: 200	4	9	~~200~~ 0
10: 50 →	5: 100 →	8: 25	~~175~~ ~~125~~ ~~250~~
12	7	6: 25	77
		50	50
~~250~~ ~~50~~ 0	~~100~~ 0	~~150~~ ~~125~~ ~~50~~ 0	

$u \backslash v$	13	8	11
0	13: 200	4: $\boxed{4}$	9: $\boxed{2}$
-3	10: 50 →	5: 100 →	8: 25
-5	12: $\boxed{-4}$	7: $\boxed{-4}$	6: 25
-11	$\boxed{2}$	$\boxed{-3}$	50

$u \backslash v$	13	8	11
0	13: $-\theta$ 200	4: θ $\boxed{4}$	9: $\boxed{2}$
-3	10: $+\theta$ 50 →	5: $-\theta$ 100 →	8: 25
-5	12: $\boxed{-4}$	7: $\boxed{-4}$	6: 25
-11	$\boxed{2}$	$\boxed{-3}$	50

$\theta = 100$

E.V. $= X_{12}$

L.V. $= X_{22}$

$u \backslash v$	13	4	11	
0	13: 100	4: 100	9: $\boxed{2}$	200
-3	10: 150	5: -4	8: 25	175
-5	12: -4	7: -8	6: 75	75
-11	2	-7	50	

$u \backslash v$	13	4	11	
0	13: $-\theta$ 100	4: 100	9: θ $\boxed{2}$	200
-3	10: $+\theta$ 150	5: -4	8: 25 $-\theta$	175
-5	12: -4	7: -8	6: 75	75
-11	2	-7	50	

$\theta = 25$

E.V. $= X_{13}$

L.V. $= X_{23}$

$u \backslash v$	13	4	9
0	13: 75	4: 100	9: 25
-3	10: 175	5: -4	8: -2
-3	12: -2	7: -6	6: 75
-9	4	-5	50

$u \backslash v$	13	4	9
0	13: $-\theta$ 75	4: 100	9: $+\theta$ 25
-3	10: 175	5: -4	8: -2
-3	12: -2	7: -6	6: 75
-9	θ 4	-5	50 $-\theta$

$\theta = 50$

E.V. $= X_{41}$

L.V. $= X_{43}$

$u \backslash v$	13	4	9
0	13: 25	4: 100	9: 75
-3	10: 175	5: -4	8: -2
-3	12: -2	7: -6	6: 75
-13	50	-9	-4

$$Z_{opt} = 25.13 + 4.100 + 9.75 + 175.10 + 75.6$$

$$= 3600$$